2.1 Quadrotor Kinematics

Transformations

$g(p)$ 作用于点 p, point p = point q

$g'(\cdot)$ 作用于向量 vector $\hat{v}$ = vector $\hat{u}$

在刚体变换中, 任何一对点, 间隔距离不变

$\|g(p)-g(q)\| = \|p-q\|$ · Length must be preserved

$g'(u) \times g'(v) = g'(u \times v)$ 刚体作用后的叉积和是作用后的结果

· Cross product are preserved

· Orthogonal vectors are mapped to orthogonal vectors

· g' preserves inner products $g'(u) \cdot g'(v) = u \cdot v$

Rigid body displacements and rigid body transformations are used interchangeably

Properties of a Rotation Matrix

· 正交

· 行列式 = 1

· 任意两个R的乘积得到一个新R

· 逆矩阵也是R

Rotations

· The group of rotations is called SO(3)

· Coordinates for SO(3)

1. Rotation matrices
2. Euler Angle
3. Axis angle representation
4. Exponential coordinates
5. Quaternions

2 is the minimum number of charts required to cover the circle's surface to create a 1 to 1 mapping

Euler Angles

$^{a}R_b = {}^{a}R_c \cdot {}^{c}R_b$

$Rot(z,\theta)$ $Rot(y,\theta)$ $Rot(z,\psi)$ linearly independent

与x,y相关性是奇数

But $\theta=0$ is a special (singular case) 而且 x 与 z 是 not linearly independent

$$R = Rot(z,\phi) \times Rot(y,\theta) \times Rot(z,\psi)$$

$$R = \begin{bmatrix} c\phi c\theta c\psi - s\phi s\psi & -c\phi c\theta s\psi - s\phi c\psi & c\phi s\theta \\ s\phi c\theta c\psi + c\phi s\psi & -s\phi c\theta s\psi + c\phi c\psi & s\phi s\theta \\ -s\theta c\psi & s\theta s\psi & \boxed{c\theta} \end{bmatrix}$$

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & \boxed{R_{33}} \end{bmatrix} \quad R_{33} = \cos\theta$$

If R_{33} is 1 or -1,

Infinite set of Euler Angles

Axis / Angle Representations for Rotations

任何一个物体绕过原点, 使得其任何一点与一点o保持不变, 都可以使其绕某轴转动

$q = Rp$ $Rp = \lambda p$ $\lambda = 1$ is an eigenvalue for any R

Rotation of a generic vector p about u through θ

$Rp = p\cos\theta + uu^T(1-\cos\theta)p + \hat{u}p\sin\theta$

$Rot(u,\theta) = I\cos\theta + uu^T(1-\cos\theta) + \hat{u}\sin\theta$ $\hat{u}$是u的反对称矩阵 (Rodrigues' formula)

已知R求 θ 和 u

$\cos\theta = \frac{Tr}{2}$ $\hat{u} = \frac{1}{2\sin\theta}(R-R^T)$

1 (axis, angle) to rotation matrix map is many to 1

2 restricting angle to the interval $[0,\pi]$ makes it 1-1 except for

$T=-1 \Rightarrow \theta=0$ no unique axis

$T=1 \Rightarrow \theta=\pi$ u or $-u$

Angular velocity

$\dot{R}^T(t)R(t) = I$ $\xrightarrow{\frac{d}{dt}(\cdot)}$ $\dot{R}^TR + R^T\dot{R} = 0$ $R^T\dot{R}$ and $\dot{R}R^T$ are skew symmetric

$R(t)R^T(t) = I$ $R\dot{R}^T + \dot{R}R^T = 0$

$q(t) = R(t)p$ $\dot{q} = \dot{R}p$ $R^T\dot{q} = R^T\dot{R}p$ encodes angular velocity in body-fixed frame $(\hat{\omega}^b)$

velocity in body-fixed frame

$\dot{q} = \dot{R}R^Tq$ inertial frame $(\hat{\omega}^s)$

Two Rotations.

$R = R_1(\theta)R_2(\phi)$

$\hat{\omega}^b = R^T\dot{R} = (R_1R_2+R_1\dot{R}_2)^T(\dot{R}_1R_2+R_1\dot{R}_2) = R_2^TR_1^T\dot{R}_1R_2 + R_2^T\dot{R}_2$

$\hat{\omega}^s = \dot{R}R^T = (\dot{R}_1R_2+R_1\dot{R}_2)(R_1R_2)^T = \dot{R}_1R_1^T + R_1\dot{R}_2R_1^T$

Quaternions

$\theta, u \rightarrow q = (\cos(\frac{\theta}{2}), u\sin(\frac{\theta}{2}), u_2\sin(\frac{\theta}{2}), u_3\sin(\frac{\theta}{2}))$

$\phi = 2\cos^{-1}(q_0)$ $u = [\frac{q_1}{\sqrt{1-q_0^2}}, \frac{q_2}{\sqrt{1-q_0^2}}, \frac{q_3}{\sqrt{1-q_0^2}}]^T$

$p = (0, \vec{p})$ $p' = qpq^{-1} = (0, \vec{p}')$ $q = q_1q_2$

Skew-Symmetric Matrices

$A^T = -A$ 对角线为0

Hat Operator

$u \times v = \hat{u}v$

2.3 Quadrotor Dynamics

Formulation

$F = F_1 + F_2 + F_3 + F_4 - mg\hat{k}$

$M = r_1\times F_1 + r_2\times F_2 + r_3\times F_3 + r_4\times F_4 + \sum_{i=1}^{4} m$

Newton-Euler Equations

$F = ma$

Center of mass $r_c = \frac{1}{m}\sum_{i=1}^{n} m_i r_i$ $F = \sum_{i=1}^{n} m\frac{d^2r_i}{dt}$

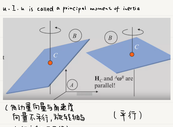
Rotational equations of motion for a rigid body

$\frac{d\,{}^{a}H_b}{dt} = M_b^a$ 角动量=转矩+加速度 ${}^{a}H_b^b = I_b\cdot{}^{a}\omega_b^b$

Principal Axes and Principal Moments of Inertia

u is a unit vector along a principal axis if $I\cdot u$ is parallel to u. 主惯性轴

$u\cdot I\cdot u$ is called a principal moment of inertia



(角动量向量与角速度向量不等时, 旋转轴总作与主轴方向相)

(平行)

Euler's Equation

$\frac{{}^{a}dH_b}{dt} = M_b$ 惯性参考系 $\vec{u}_1, \vec{u}_2, \vec{u}_3$ 为主轴

Let $\vec{u}_1, \vec{u}_2, \vec{u}_3$ be along principal axes and ${}^{a}\omega^b = \omega_1\vec{u}_1 + \omega_2\vec{u}_2 + \omega_3\vec{u}_3$

因为主轴是固定的

$\Rightarrow \frac{d\,{}^{b}H_b}{dt} + {}^{a}\omega^b \times H_b = M_b$

$\frac{{}^{b}dH_b}{dt} = I_{11}\dot{\omega}_1\vec{u}_1 + I_{22}\dot{\omega}_2\vec{u}_2 + I_{33}\dot{\omega}_3\vec{u}_3$

Quadrotor Equations of Motion

${}^{a}\omega^b = p\vec{u}_1 + q\vec{u}_2 + r\vec{u}_3$

$\omega = \begin{bmatrix} p \\ q \\ r \end{bmatrix} = R\begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix}$

$I\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} + \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times I\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} l(F_1-F_3) \\ l(F_2-F_4) \\ C_M(-F_1+F_2-F_3+F_4) \end{bmatrix}$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} c\theta & 0 & -c\phi s\theta \\ 0 & 1 & s\phi \\ s\theta & 0 & c\phi c\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \begin{matrix} \text{Pitch} \\ \text{Roll} \\ \text{Yaw} \end{matrix}$$

Body frame 角速度

State Space for Quadrotor

$q = \begin{bmatrix} x \\ y \\ z \\ \phi \\ \theta \\ \psi \end{bmatrix}$ $x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$ q: 位置 $\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}$ (姿态)

Planner Quadrotor Model

$m\ddot{y} = \sin\theta(\phi)u_1$

$m\ddot{z} = \cos(\theta)u_1 + mg$

$I_{xx}\ddot{\phi} = u_2$

1. 平面
2. Defines states $x_1 = y$, $x_2 = \dot{z}$, $x_3 = \phi$

$x_4 = y$, $x_5 = z$, $x_6 = \dot{\phi}$

3. State vector $x = [x_1, x_2, x_3, x_4, x_5, x_6]^T = [y, \dot{z}, \phi, y, z, \dot{\phi}]^T$

4. $\frac{d}{dt}x_1 = \frac{d}{dt}y = y = x_4$ $\frac{d}{dt}x_2 = \frac{d}{dt}\dot{z} = \ddot{z} = x_5$ $\frac{d}{dt}x_3 = \frac{d}{dt}\phi = \dot{\phi} = x_6$

$\frac{d}{dt}x_4 = \frac{d}{dt}y = \ddot{y} = \frac{\sin(\phi)u_1}{m}$, $\frac{\sin(\phi)u_1}{m} = \frac{\sin(x_3)u_1}{m}$ $\frac{d}{dt}x_5 = \frac{d}{dt}\dot{z} = \ddot{z} = \frac{\cos(\phi)u_1}{m} - g$ $\frac{d}{dt}x_6 = \frac{d}{dt}\dot{\phi} = \dot{\phi} = \frac{u_2}{I_{xx}}$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_5 \\ x_6 \\ \frac{\sin(x_3)u_1}{m} \\ \frac{\cos(x_3)u_1}{m} - g \\ \frac{u_2}{I_{xx}} \end{bmatrix}$$